Nearest-Neighbor Escape Probability of SRW on $\mathbb{Z}^2$

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Problem

Consider a simple random walk on $\mathbb{Z}^2$ starting at the origin $(0,0)$. What is the probability that it hits $(1,0)$ before it returns to $(0,0)$?

In probability notation, we want

$$\mathbb{P}\big(\tau_{(1,0)} < \tau^+_{(0,0)} \,\big|\, X_0 = (0,0)\big),$$

where

$$\tau_{(1,0)} = \inf\{t \geq 0 : X_t = (1,0)\}, \qquad \tau^+_{(0,0)} = \inf\{t > 0 : X_t = (0,0)\}.$$

Equivalently, this is the escape probability of the random walk from $(0,0)$ to $(1,0)$ before returning.

Electric Network Interpretation

Consider the infinite network $\mathbb{Z}^2$ with unit resistance on each edge. If we can compute the effective resistance between the nodes $(0,0)$ and $(1,0)$, then the escape probability is given by

$$\mathbb{P}\big(\tau_{(1,0)} < \tau^+_{(0,0)}\big) = \frac{1}{4R_{\text{eff}}((0,0),(1,0))}.$$

So the whole problem reduces to proving

$$R_{\text{eff}}((0,0),(1,0)) = \frac{1}{2}.$$

Two-Flow Construction on the Infinite Network

Let

$$o = (0,0), \qquad a = (1,0).$$

At first sight, one wants to consider two networks:

- **Network 1:** Put 1A of current into o and take 1A of current out from "infinitely far away".

- **Network 2:** Put 1A of current into the "infinitely far away" and take 1A of current out from a.

If one were allowed to work with these two infinite networks directly, then the answer would look immediate.

For the first network, by symmetry at o, the current splits equally into the four adjacent edges, so the current through the edge

$$o \to a$$

should be

$$I^{(1)}_{o\to a} = \frac{1}{4}.$$

For the second network, by symmetry at a, the current should also split equally among the four edges adjacent to a, so the current through the same edge should again be

$$I^{(2)}_{o\to a} = \frac{1}{4}.$$

Now if one adds the two networks, one gets a new network with 1A entering at o and 1A leaving at a. By linearity of electric networks, the current through the edge $o \to a$ should then be

$$I_{o\to a} = I^{(1)}_{o\to a} + I^{(2)}_{o\to a} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Since the edge resistance is 1, this would give

$$V(o) - V(a) = 1 \cdot I_{o\to a} = \frac{1}{2},$$

hence

$$R_{\text{eff}}(o,a) = \frac{1}{2}, \qquad \mathbb{P}\big(\tau_a < \tau_o^+\big) = \frac{1}{4R_{\text{eff}}(o,a)} = \frac{1}{2}.$$

This argument is very natural, but there is a real problem here: the phrase "infinitely far away" has not been defined, and in a recurrent graph like $\mathbb{Z}^2$ one cannot simply superpose two infinite networks without justification.

So the real question is how to make this idea rigorous.

Finite-Box Approximation

Let

$$B_n^{(0)} = [-n,n]^2 \cap \mathbb{Z}^2.$$

Wire all points of $\partial B_n^{(0)}$ into a single boundary vertex.

We first define two finite networks on this wired box.

- **Network 1 on $B_n^{(0)}$:** put 1A into o and take 1A out from the wired boundary.
- **Network 2 on $B_n^{(0)}$:** put 1A into the wired boundary and take 1A out from a.

Let

$$i_n^{(1)} \quad \text{be the current of Network 1 on } B_n^{(0)},$$

and

$$i_n^{(2)} \quad \text{be the current of Network 2 on } B_n^{(0)}.$$

Their sum

$$\widetilde{i}_n := i_n^{(1)} + i_n^{(2)}$$

is exactly the unit current flow from o to a on the wired box $B_n^{(0)}$.

So the philosophy is exactly the same as before. The only issue is now to justify what happens as $n \to \infty$.

Network 1

For Network 1, the box $B_n^{(0)}$ is centered at o, so the four edges adjacent to o are symmetric. Therefore the current leaving o splits equally among these four edges. In particular,

$$i_n^{(1)}(o, a) = \frac{1}{4} \qquad \text{for every } n.$$

Network 2

The second network is harder because the box $B_n^{(0)}$ is not centered at a. So even though in the infinite picture one expects symmetry around a, at finite n there is no exact symmetry forcing

$$i_n^{(2)}(o, a) = \frac{1}{4}.$$

This is the point where my original argument was incomplete.

A wrong first attempt is to consider

$$v_n(x) := \mathbb{P}_x(\tau_{\partial B_n^{(0)}} < \tau_a),$$

which is the unit voltage function with boundary value 1 at $\partial B_n^{(0)}$ and 0 at a. But since simple random walk on $\mathbb{Z}^2$ is recurrent, for each fixed x we have

$$v_n(x) \to 0 \qquad (n \to \infty),$$

so this normalization is useless for the present problem.

The correct normalization is to multiply by the effective resistance from a to the boundary.

Normalized Voltage for Network 2

Let

$$r_n := R_{\text{eff}}\big(a \leftrightarrow \partial B_n^{(0)}\big).$$

Define

$$V_n^{(0)}(x) := r_n\, v_n(x).$$

Then $V_n^{(0)}$ is the voltage function for Network 2 on the box $B_n^{(0)}$, normalized by

$$V_n^{(0)}(a) = 0.$$

Equivalently, $V_n^{(0)}$ satisfies

$$V_n^{(0)}(a) = 0,$$

$$\Delta V_n^{(0)}(x) = 0 \qquad \text{for } x \in B_n^{(0)} \setminus \{a\},$$

and the total current extracted at a is 1, namely

$$\sum_{y\sim a} \big(V_n^{(0)}(y) - V_n^{(0)}(a)\big) = 1.$$

Since $V_n^{(0)}(a) = 0$, this becomes

$$\sum_{y\sim a} V_n^{(0)}(y) = 1.$$

If one could prove that, for each fixed x,

$$V_n^{(0)}(x) \longrightarrow V(x),$$

and that this limit agrees near a with the symmetric limit coming from a box centered at a, then necessarily the four neighbors of a would have the same limit and sum to 1, hence each limit would be $1/4$. That would give exactly what we want.

A Symmetric Comparison Network

Define the box centered at a by

$$B_n^{(a)} := ([-n+1, n+1] \times [-n, n]) \cap \mathbb{Z}^2.$$

Again wire $\partial B_n^{(a)}$ into a single boundary vertex.

Define $V_n^{(a)}$ to be the voltage for the unit current flow from the wired boundary of $B_n^{(a)}$ to a, normalized by

$$V_n^{(a)}(a) = 0.$$

So $V_n^{(a)}$ satisfies

$$V_n^{(a)}(a) = 0,$$

$$\Delta V_n^{(a)}(x) = 0 \qquad \text{for } x \in B_n^{(a)} \setminus \{a\},$$

and

$$\sum_{y\sim a} \big(V_n^{(a)}(y) - V_n^{(a)}(a)\big) = 1.$$

Since the box $B_n^{(a)}$ is centered at a, the four neighbors of a are symmetric, hence

$$V_n^{(a)}(0,0) = V_n^{(a)}(2,0) = V_n^{(a)}(1,1) = V_n^{(a)}(1,-1) = \frac{1}{4}.$$

In particular,

$$i_{n,\mathrm{sym}}^{(2)}(o,a) = V_n^{(a)}(o) - V_n^{(a)}(a) = \frac{1}{4}.$$

So the remaining problem is:

Show that, for fixed x near a,

$$V_n^{(0)}(x) - V_n^{(a)}(x) \to 0?$$

If yes, then

$$V_n^{(0)}(o) \to \frac{1}{4},$$

therefore

$$i_n^{(2)}(o,a) = V_n^{(0)}(o) - V_n^{(0)}(a) \to \frac{1}{4}.$$

Green Function Representation

For a finite set $D \subset \mathbb{Z}^2$, let

$$\tau_{\partial D} := \inf\{k \geq 0 : X_k \notin D^\circ\},$$

and define the killed Green function

$$g_D(x,y) := \mathbb{E}_x\left[\sum_{k=0}^{\tau_{\partial D}-1} \mathbf{1}_{\{X_k = y\}}\right].$$

For the unit-resistance network, the voltage of the unit current flow from a to ∂D is

$$U_D(x) = \frac{1}{4}\, g_D(x,a).$$

Therefore the voltage of the reversed unit current flow from ∂D to a, normalized by $V_D(a) = 0$, is

$$V_D(x) := U_D(a) - U_D(x) = \frac{1}{4}\big(g_D(a,a) - g_D(x,a)\big).$$

Hence

$$V_n^{(0)}(x) = \frac{1}{4}\big(g_{B_n^{(0)}}(a,a) - g_{B_n^{(0)}}(x,a)\big),$$

and

$$V_n^{(a)}(x) = \frac{1}{4}\big(g_{B_n^{(a)}}(a,a) - g_{B_n^{(a)}}(x,a)\big).$$

So the comparison between the two voltage functions is exactly a comparison between two killed Green functions.

Potential Kernel Formula

$$a(z) = \frac{2}{\pi}\log|z| + c_0 + O(|z|^{-2}) \qquad (|z| \to \infty).$$

Standard representation of the killed Green function:

$$g_D(x,a) = \mathbb{E}_x\big[\mathfrak{a}(X_{\tau_{\partial D}} - a)\big] - \mathfrak{a}(x-a).$$

Substituting this into the formula for V_D, we obtain

$$V_D(x) = \frac{1}{4}\Big(\mathbb{E}_a[\mathfrak{a}(X_{\tau_{\partial D}} - a)] - \mathbb{E}_x[\mathfrak{a}(X_{\tau_{\partial D}} - a)] + \mathfrak{a}(x-a)\Big).$$

Apply this formula to the two domains $D = B_n^{(0)}$ and $D = B_n^{(a)}$.

Comparison of the two boxes

Let

$$D_0 := B_n^{(0)}, \qquad D_a := B_n^{(a)}.$$

Recall that

$$V_D(x) = \frac{1}{4}\left(\mathbb{E}_a\big[\mathfrak{a}(X_{\tau_D} - a)\big] - \mathbb{E}_x\big[\mathfrak{a}(X_{\tau_D} - a)\big] + \mathfrak{a}(x-a)\right).$$

Therefore

$$\begin{aligned} V_{D_0}(x) - V_{D_a}(x) = \frac{1}{4}\Big(&\mathbb{E}_a\big[\mathfrak{a}(X_{\tau_{D_0}} - a)\big] - \mathbb{E}_a\big[\mathfrak{a}(X_{\tau_{D_a}} - a)\big] \\ &- \mathbb{E}_x\big[\mathfrak{a}(X_{\tau_{D_0}} - a)\big] + \mathbb{E}_x\big[\mathfrak{a}(X_{\tau_{D_a}} - a)\big]\Big). \end{aligned}$$

$\mathfrak{a}(x-a)$ has disappeared, since it does not depend on the domain.

For a domain D, write the exit distribution as

$$H_D(y,\xi) := \mathbb{P}_y(X_{\tau_D} = \xi), \qquad \xi \in \partial D.$$

Then

$$\mathbb{E}_y\big[\mathfrak{a}(X_{\tau_D} - a)\big] = \sum_{\xi\in\partial D} H_D(y,\xi)\,\mathfrak{a}(\xi - a).$$

Hence

$$\begin{aligned} V_{D_0}(x) - V_{D_a}(x) = \frac{1}{4}\Bigg[&\sum_{\xi\in\partial D_0} H_{D_0}(a,\xi)\,\mathfrak{a}(\xi-a) - \sum_{\eta\in\partial D_a} H_{D_a}(a,\eta)\,\mathfrak{a}(\eta-a) \\ &- \sum_{\xi\in\partial D_0} H_{D_0}(x,\xi)\,\mathfrak{a}(\xi-a) + \sum_{\eta\in\partial D_a} H_{D_a}(x,\eta)\,\mathfrak{a}(\eta-a)\Bigg]. \end{aligned}$$

Now D_0 and D_a differ only by a shift of the boundary by one lattice unit. Thus the relevant boundary points can be paired as

$$\eta = \xi + u, \qquad |u| \le 1.$$

Moreover, all these boundary points satisfy

$$|\xi - a| \asymp n, \qquad |\eta - a| \asymp n.$$

Using the potential kernel asymptotic

$$\mathfrak{a}(z) = \frac{2}{\pi} \log |z| + c_0 + O(|z|^{-2}),$$

we get, uniformly for $|z| \asymp n$ and $|u| \le 1$,

$$\mathfrak{a}(z+u) - \mathfrak{a}(z) = O\left(\frac{1}{n}\right).$$

$$\mathbb{E}_y\big[\mathfrak{a}(X_{\tau_{D_0}} - a)\big] - \mathbb{E}_y\big[\mathfrak{a}(X_{\tau_{D_a}} - a)\big] = O\left(\frac{1}{n}\right),$$

for every fixed starting point y in a bounded neighborhood of a. Applying this with $y = a$ and $y = x$, we obtain

$$V_{D_0}(x) - V_{D_a}(x) = O\left(\frac{1}{n}\right).$$

In particular, taking $x = o$,

$$V_n^{(0)}(o) - V_n^{(a)}(o) = O\left(\frac{1}{n}\right) \to 0.$$

But by the local current computation,

$$V_n^{(a)}(o) = \frac{1}{4}.$$

Hence

$$V_n^{(0)}(o) \to \frac{1}{4}.$$

Since $V_n^{(0)}(a) = 0$, it follows that

$$i_n^{(2)}(o, a) = V_n^{(0)}(o) - V_n^{(0)}(a) \to \frac{1}{4}.$$

Passing to the Limit

Now we know:

- for every n,

$$i_n^{(1)}(o, a) = \frac{1}{4};$$

- and

$$i_n^{(2)}(o, a) \to \frac{1}{4}.$$

Since

$$\widetilde{i}_n = i_n^{(1)} + i_n^{(2)}$$

is the unit current flow from o to a on the wired box $B_n^{(0)}$, the current through the edge $o \to a$ satisfies

$$\widetilde{i}_n(o,a) = i_n^{(1)}(o,a) + i_n^{(2)}(o,a) \to \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Because o and a are adjacent and the edge resistance is 1, the voltage drop across that edge is

$$V_n(o) - V_n(a) = \widetilde{i}_n(o,a),$$

hence

$$R_{\mathrm{eff}}^{B_n^{(0)}}(o,a) \to \frac{1}{2}.$$

By wired exhaustion,

$$R_{\mathrm{eff}}^{\mathbb{Z}^2}(o,a) = \lim_{n\to\infty} R_{\mathrm{eff}}^{B_n^{(0)}}(o,a) = \frac{1}{2}.$$

Therefore

$$\mathbb{P}\big(\tau_a < \tau_o^+\big) = \frac{1}{4R_{\mathrm{eff}}^{\mathbb{Z}^2}(o,a)} = \frac{1}{4\cdot(1/2)} = \frac{1}{2}.$$